

DFMEA - Design L (Case T2R1NOCEILING)

Top risks FM1/FM2 are boundary-level (material system / electrode modification locked, escalation OFF).

FM	Failure mode	Effect / cause	S O D	RPN	Status
1	4C lithium plating	Test start state + SPMe surface response + electrolyte temp	10 10 2	200	Out of admissible scope
2	SEI500 overgrowth (811Pe-550)	4.2 V charge-end depth	8 10 2	160	Out of admissible scope
3	SEI100 erosion (thick and dense)	at 540.0 nm (R3)	7 3 1	21	Direction rejected
4	Near-zero 4C acceptance	0.0461 Ah in 8.4 s (~0.9% SOC)	7 10 2	140	System-level limitation
5	lowT margin vs cooling	h 100 -> 0.9919 (l)	4 3 1	12	Keep h 10
6	4C thermal rise	333.2 K Ohmic heating in 4C burst	3 8 2	48	Monitor
7	ED overestimate (no electrolyte)	Only at formula excludes electrolyte	4 10 1	40	Annotated